

三、黎曼曲率张量

(无挠) 导数算符: $\nabla: \mathcal{F}_M(k, l) \rightarrow \mathcal{F}_M(k, l+1)$ (∇ 在 M 上)

derivative operator

1) 线性 2) Leibnitz 律: $\nabla_a T^{b_1 \dots b_k}_{c_1 \dots c_l} S^{d_1 \dots d_k}_{e_1 \dots e_k} = T \nabla_a S + S \nabla_a T$

3) 与缩并可换序 $\nabla \cdot c = c \cdot \nabla$ (eg. $\nabla_a (v^b w_b) = v^b \nabla_a w_b + w_b \nabla_a v^b$) (等价: $\nabla_a \delta^b_c = 0$)

4) $v(f) = v^a \nabla_a f$, $\forall f \in \mathcal{F}_M, v \in \mathcal{F}_M(1, 0)$

proof: $\Rightarrow \nabla_a v^b = \nabla_a (\delta^b_c v^c) = v^c \nabla_a \delta^b_c + \nabla_a v^b$

5) 无挠性: $\nabla_a \nabla_b f = \nabla_b \nabla_a f$, $f \in \mathcal{F}_M$ (标函)
torsion free

$\Leftarrow \nabla_a$ 满足 1~4), $\nabla_a f = \tilde{\nabla}_a f = df$

局域性: T 在 $p \in M$ 的邻域 U , $\nabla_a T$ 仅在 U 上合法
 $\downarrow$
(U 内 $\nabla_a \bar{T}|_U = \nabla_a T|_U$)
拓展 ∇ 在 M 上

对函数 f , $\nabla_a f = \tilde{\nabla}_a f = (df)_a$

对 $(0, 1)$ 张量 w_b 与 w'_b , $\nabla_a w'_b \neq \nabla_a w_b$ ($w'_b|_p = w_b|_p$)

proof: $\Omega_b = w'_b - w_b = \Omega_M(dx^a)_b$

但是 $\star$: 对两个导数算符: $(\nabla_a - \tilde{\nabla}_a)w_b|_p = (\nabla_a - \tilde{\nabla}_a)w'_b|_p$
或 $\nabla_a (w'_b - w_b)|_p = \tilde{\nabla}_a (w'_b - w_b)|_p$

$\leftarrow \nabla_a \Omega_M = \tilde{\nabla}_a \Omega_M = (d\Omega_M)_a$

$\nabla_a w_b|_p$ 与 p 邻域有关 $\leftarrow$

对无挠算符 ∇ 与 $\tilde{\nabla}$, $C^c_{ab} = C^c_{ba}$

但 $(\tilde{\nabla}_a - \nabla_a)w_b$ 仅与 p 点有关 $\rightarrow$ 定义: $(\tilde{\nabla}_a - \nabla_a)w_b = C^c_{ab} w_c$, 即: $\nabla_a w_b = \tilde{\nabla}_a w_b - C^c_{ab} w_c$

逆+协: 张量: $T \in \mathcal{F}_M(k, l)$ (逆) $\nabla_a v^b = \tilde{\nabla}_a v^b + C^b_{ac} v^c$

$$\nabla_a T^{b_1 \dots b_k}_{c_1 \dots c_l} = \tilde{\nabla}_a T^{b_1 \dots b_k}_{c_1 \dots c_l} + \sum_i C^{b_i}_{ad} T^{b_1 \dots b_k}_{c_1 \dots c_l} - \sum_j C^{c_j}_{ad} T^{b_1 \dots b_k}_{c_1 \dots c_l}$$

(注意 $\nabla_a T^b_c = T^b \nabla_a T_c + T_c \tilde{\nabla}_a T_b$)

$\nabla_a (w_b v^b) = \tilde{\nabla}_a (w_b v^b)$

$w_b \nabla_a v^b + v^b \nabla_a w_b = w_b \tilde{\nabla}_a v^b + v^b \tilde{\nabla}_a w_b$

$w_b (\nabla_a v^b - C^b_{ac} v^c) = \tilde{\nabla}_a (w_b v^b)$, proven

定义: $\partial_a: \mathcal{F}(k,l) \rightarrow \mathcal{F}(k,l+1)$, $\partial_a T^b{}_c := (dx^a)_a (\partial/\partial x^b)^b (dx^c)_c \cdot \underbrace{\partial_\mu T^\nu}_\frac{\partial}{\partial x^\mu}$

普通导数算符:

ordinary derivative operator

$\partial_a \uparrow$

性质: 1) 不与自己基矢作用

$$\partial_a (\partial/\partial x^b)^b = \partial_a (dx^b)_b = 0$$

2) 无挠

$$\partial_a \partial_b T = \partial_b \partial_a T \quad (\partial_{[a} \partial_{b]} T = 0) \quad T \in \mathcal{F}_M(k,l)$$

与 ∇_a 的无挠相比, ∇_a 对 $\mathcal{F}_M(0,0)$ ∂_a 对 $\mathcal{F}_M(k,l)$ 都可

协变导数算符: ∇_a 与坐标系无关

Christoffel symbol 克氏符 $\Gamma^c{}_{ab}$ 为 ∇_a 与 ∂_a 的差 $C^c{}_{ab}$

$\therefore \Gamma^c{}_{ab}$ 可叫作张量, 本书 $\Gamma^c{}_{ab}$ 定义为 $\nabla_a - \partial_a$, ∂_a 与坐标有关, 故为张量。一些书定义 $\Gamma^b{}_{ac}$ ($\Gamma^a{}_{bc}$ 在 $\{x^a\}$ 的分量) 为克氏符, 故不看作张量
坐标变换满足张量变换规则

$$\partial_a V^b = (dx^a)_a (\partial/\partial x^b)^b V^c{}_{,a} \quad (V^c{}_{,a} \equiv \partial_a V^c)$$

$$\text{简记: } V^c{}_{,a} = \partial_a V^c, \quad V^c{}_{;a} = \nabla_a V^c$$

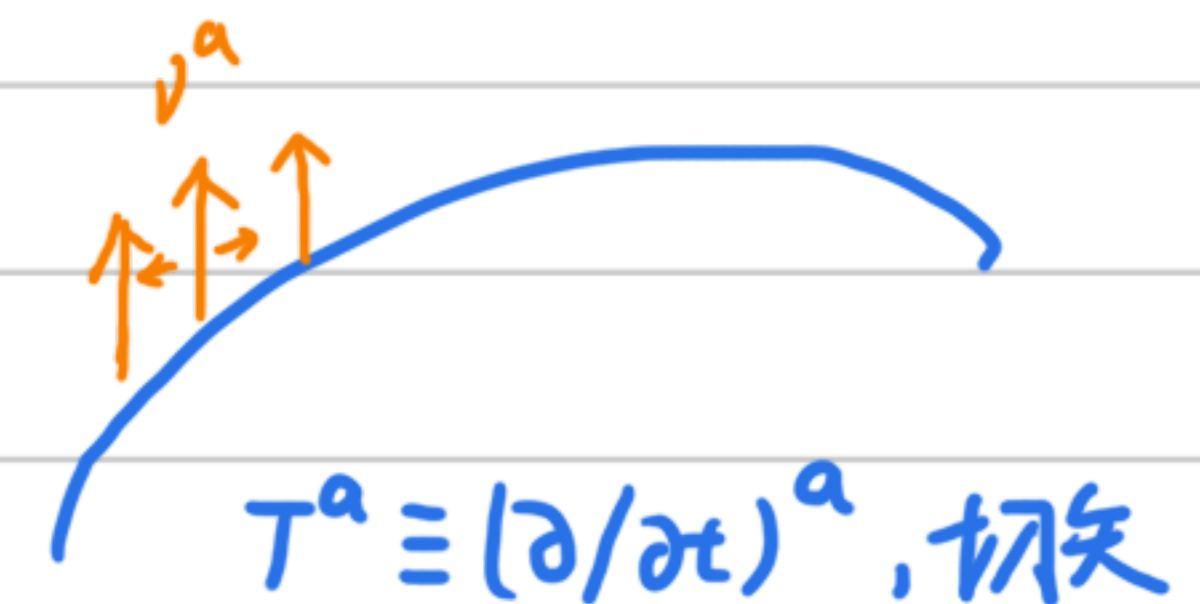
$$\downarrow \quad V^c{}_{;a} = V^c{}_{,a} + \Gamma^c{}_{ab} V^b, \quad W_{c;a} = W_{c,a} - \Gamma^b{}_{ac} W_b$$

矢量场的平移

parallelly transported
along $C(t)$ 平移的

若 V^a 矢量场 沿着曲线 $C(t)$, $T^b \nabla_b V^a = 0$
(平移不发生变化)

协导



$$T^b \nabla_b V^a = (\partial/\partial x^u)^a (dv^u/dt + \Gamma^u_{\nu\delta} T^\nu V^\delta)$$

正常而言, V^b 定义在 $C(t)$ 上, $\nabla_b V^a$ 无意义, 但前有 T^b , 无意义项为 0, 故可以
(可作一个小的延拓, V^b 变为 $\bar{V}^b$, $\bar{V}^b$ 在邻域有定义)

曲线上一点 $C(t_0)$ 及该点一个矢量 唯一决定沿曲平移的矢量场

proof: 对一个坐标域内: $T^b \nabla_b V^a = 0 \Rightarrow dv^u/dt + \Gamma^u_{\nu\delta} T^\nu V^\delta = 0$

多个坐标域: 接力法 (邻域 U : 初态 $\rightarrow$ 确定 $\rightarrow$ 解微分方程 $\rightarrow$ 终态 (接力))
? $\square$ 空间解释: 微小坐标基 $\rightarrow$ 近似线性 $\rightarrow$ 本征向量不变?

proof:

$$\nabla_b V^a = \frac{\partial_a V^a}{(dx^b)^b (\partial/\partial x^u)^a \cdot \partial_v x^u} + \Gamma^a_{bc} V^c$$

$T^b = dx^b(t)/dt$, 代入:

$$T^b \nabla_b V^a = (\partial/\partial x^u)^a \cdot T^b (\partial V^u / \partial x^b) + (\partial/\partial x^u)^a \Gamma^u_{\nu\delta} T^\nu V^\delta = (\partial/\partial x^u)^a \cdot ((dv^u/dt) + \Gamma^u_{\nu\delta} T^\nu V^\delta)$$

联络 $v^a \in V_p, v^b \in V_q, \exists$ 线 $C(t)$ 连接 p, q , $C(t)$ 上的唯一平移矢量场 起点为 v^a , 终点为 $\bar{v}^b$
connection $\hookrightarrow$ 映射 $\nabla_a := v^a \rightarrow \bar{v}^b$

与度规适配的导数算符 取 C 上平移的 v^a, u^b $\star$

$$u = v^a g_{ab} v^b$$

核心: $T^c \nabla_c U = 0$ (内积不随平移变化)

$$0 = T^c \nabla_c (g_{ab} v^a u^b) = g_{ab} T^c (\underbrace{u^b \nabla_c v^a + v^a \nabla_c u^b}_{\nabla_c u^b v^a = 0 ?}) + v^a u^b T^c \nabla_c (g_{ab})$$

$$\nabla_c u^b v^a = 0 ? \square$$

$$\nabla_c g_{ab} = 0$$

当确定度规时, 存在唯一的 ∇_a 使 $\nabla_a g_{bc} = 0$

proof: $\forall \tilde{\nabla}_a$, 有:

$$\nabla_a g_{bc} = \tilde{\nabla}_a g_{bc} - C^d_{ab} g_{dc} - C^d_{ac} g_{bd} = 0$$

$$\hookrightarrow \tilde{\nabla}_a g_{bc} = \underbrace{C^d_{ab} g_{dc}}_{\text{记为 } C_{cab}} + \underbrace{C^d_{ac} g_{bd}}_{C_{bac}}$$

$g_{ab} = g_{ba}$ (已定义)

$$\tilde{\nabla}_a g_{bc} = C_{cab} + C_{bac}$$

$$\tilde{\nabla}_b g_{ac} = C_{cba} + C_{abc}$$

$$\tilde{\nabla}_c g_{ab} = C_{bca} + C_{acb}$$

$$\boxed{C_{abc} = C_{acb}}$$

$$\hookrightarrow C_{cab} = \frac{1}{2} (\tilde{\nabla}_a g_{bc} + \tilde{\nabla}_b g_{ac} - \tilde{\nabla}_c g_{ab})$$

$$C^c_{ab} = \frac{1}{2} g^{cd} (\tilde{\nabla}_a g_{bd} + \tilde{\nabla}_b g_{ad} - \tilde{\nabla}_d g_{ab})$$

$\hookrightarrow$ 可从 $\tilde{\nabla}_a$ 推到 ∇_a , 存在唯一 (唯一-易证)

若 $\tilde{\nabla}_a$ 为普通导数算符 ∂_a , 则有:

$$\Gamma^c_{ab} \text{ 分量 } \Gamma^b_{uv} = \frac{1}{2} g^{bp} (g_{pu,v} + g_{vp,u} - g_{uv,p})$$

$$\Gamma^c_{ab} = \frac{1}{2} g^{cd} (\partial_a g_{bd} + \partial_b g_{da} - \partial_d g_{ab})$$

矢量沿导数平移

定义矢量导数: $C(t)$ 上 $\vec{v}$, $\frac{d\vec{v}}{dt}|_P := \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} (\vec{\tilde{v}}|_P - \vec{v}|_P)$

$\Delta t \equiv t(Q) - t(P)$, $\vec{\tilde{v}}_P$ 为 $\vec{v}_Q$ 平移到 P 处的结果

$\hookrightarrow$ 等价于 $T^b \partial_b v^a$

测地线

geodesic

$\gamma(t)$, 切矢 T^a 满足 $T^b \nabla_b T^a = 0$ (切矢沿该线平移)

等价于 $S = \int \sqrt{T^a T^a g_{ab}} dt \min \rightarrow \frac{dT^u}{dt} + \Gamma^u_{\nu\delta} T^\nu T^\delta = 0 \xrightarrow{T^a = \frac{dx^a}{dt}} \frac{d^2 x^u}{(dt)^2} + \Gamma^u_{\nu\delta} \frac{dx^\nu}{dt} \frac{dx^\delta}{dt} = 0$

重参数化: $T'^b \nabla_b T'^a = \alpha T'^a$

proof: $T^a = \left(\frac{\partial}{\partial t}\right)^a = \left(\frac{\partial}{\partial t'}\right)^a \frac{dt'}{dt} = T'^a \cdot \frac{dt'}{dt}$

$\hookrightarrow 0 = T^b \nabla_b T^a = T^b \left(\frac{dt'}{dt}\right)^a \nabla_b T'^a + \underbrace{T'^a \frac{dt'}{dt}}_{\Rightarrow T'^a \cdot \frac{dt'}{dt} \cdot \frac{d}{dt} \left(\frac{dt'}{dt}\right)} T'^b \nabla_b \left(\frac{dt'}{dt}\right) = T'^a \cdot \frac{dt'}{dt}$

$\hookrightarrow T'^b \nabla_b T'^a = \underbrace{-\left(\frac{dt}{dt'}\right)^2 \frac{dt'}{dt}}_{\alpha} \cdot T'^a$

$\rightarrow$ 若 T^a 满足 $T^a \nabla_a T^b = \alpha T^b$, 则可找到 $t'(t)$ 使之成为测地线

$\downarrow$ affine parameter
 t' 是 γ 的仿射参数

唯一决定: 流形 M 上点 p 与 矢量 v^a 确定唯一测地线 $\gamma(t)$: 1) $\gamma(0) = p$, 2) $\gamma(t)$ 在 p 切矢为 v^a

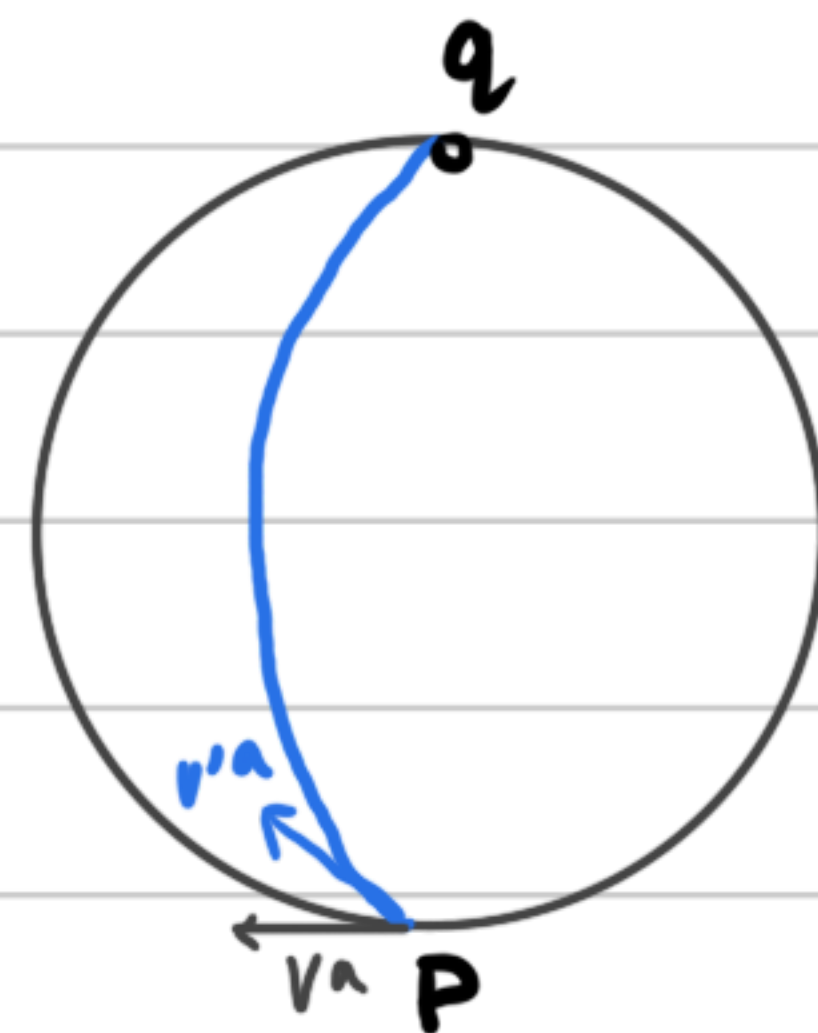
proof: 方程个数 + 条件个数

广义黎曼空间指数映射 exponential map

V_p 到流形 M 的映射 $\exp_p: V_p \rightarrow M$

$(p, v^a) \rightarrow \gamma(t)$, 定义: $e_p^{v^a} = \gamma(1)$, $e_p^{\vec{0}} = \gamma(0) = p$
(向量)

处理: 对于相交的 q , 将其它 v^a 上的 $\gamma(1)$ 挖去,
于是该映射为一一映射, 微分同胚



子群: V_p 的子群 $\hat{V}_p \rightarrow M$ 的子群 N
注: $\vec{0}$ p

黎曼法坐标

Riemannian normal coordinate 法邻域 N : $\exists V_p$ 子群 $\hat{V}_p$ 使 $\exp_p: \hat{V}_p \rightarrow N$ 微分同胚
normal neighborhood

在 N 内定义坐标: 任选一基底 $\{e_\mu^a\}$, $v^a = \exp_p^{-1}(q)$, v^a 的分量为 q 的坐标

好处: (N, ψ) 为 p 点上的法坐标:

1) $\gamma(t)$ 在 ψ 映射 $\psi(\gamma(t))$ 为直线 过 p 测地线为直线

proof: $p = \gamma(0)$, $v^a = (\partial/\partial t)^a|_p$, $q = \gamma(1)$, $q = \exp_p(v^a)$

and let $q \in \gamma(t)$, $\exists t'$ makes: $q = \gamma'(1)$, $\gamma'(t') = \gamma'(t)$, $q = \exp(v^a)$

where $v^a = (\partial/\partial t)^a|_p = (\partial/\partial t)^a|_p \cdot \frac{dt}{dt'}|_p = \alpha v_1^a$

故为直线

2) p 点克氏符为 0: $\Gamma^c_{ab}|_p = 0$

proof: $\frac{d^2 x^\mu}{dt^2} + \Gamma^\mu_{\nu\delta} \frac{dx^\nu}{dt} \frac{dx^\delta}{dt} = 0$ 为测地线 $\gamma(t)$, 这里为直线 (1), 故 $\frac{d^2 x^\mu}{dt^2} = 0$

在 p 点: $\Gamma^\mu_{\nu\delta}|_p \cdot T^\nu T^\delta = 0$, 此时切矢任意: $\Gamma^c_{ab}|_p = 0$

黎曼曲率张量

Riemann curvative tensor

对易子
commutator

∇_a 的对易子: $(\nabla_a \nabla_b - \nabla_b \nabla_a) = [\nabla_a, \nabla_b]$ 不是反对称部分

无挠: $(\nabla_a \nabla_b - \nabla_b \nabla_a)$ 作用于函数上为0 (其余张量则否)

1. 函数无关: $[\nabla_a, \nabla_b](f w_c) = f [\nabla_a, \nabla_b] w_c$

2. 单点有效 若 $w_c|_p = w'_c|_p$, 则 $[\nabla_a, \nabla_b] w_c|_p = [\nabla_a, \nabla_b] w'_c|_p$
proof: let $\Omega_c = w_c - w'_c$, $\Omega_c|_p = 0$

? $\square$

$$\nabla_a \nabla_b \Omega_c|_p = \Omega^u \nabla_a \nabla_b (\partial/\partial x^u)_c \dots \dots$$

黎曼曲率张量: $[\nabla_a, \nabla_b] w_c = R_{abc}{}^d w_d$, $\forall w_c \in \mathcal{F}(0,1)$

$\hookrightarrow$ 导数算符的非对易性

$$\nabla_a \Leftrightarrow g_{ab} \Leftrightarrow \text{空间}$$

$R_{abc}{}^d = 0 \rightarrow$ 平直度规, 可对易 e.g. $(R^i{}_j \delta_{ab}) \rightarrow \partial_a \rightarrow R_{abc}{}^d \cdot w_d = (\partial_a \partial_b - \partial_b \partial_a) w_c = (dx^u)_a (dx^v)_b (dx^d)_c (\partial_u \partial_v w_d - \partial_v \partial_u w_d) = 0$
 $(R^i{}_j \eta_{ab}) \nearrow$ 问: 伪欧几里德空间

$\hookrightarrow$ 性质: 1) $R_{abc}{}^d = -R_{bac}{}^d$ (或: $R_{(ab)c}{}^d = 0$) ($[\nabla_a, \nabla_b] = 0$)

2) $R_{[abc]}{}^d = 0$

循环恒等式
(第一比安基)

$$\text{pf: } R_{[abc]}{}^d w_d = \frac{1}{2} ([\nabla_a, \nabla_b] w_c + [\nabla_b, \nabla_c] w_a + [\nabla_c, \nabla_a] w_b - [\nabla_b, \nabla_a] w_c - [\nabla_c, \nabla_b] w_a - [\nabla_a, \nabla_c] w_b) = 2 \nabla_{[a} \nabla_b w_{c]} = 2 \nabla_a (\nabla_b w_c) - \nabla_a (\nabla_b w_c) = \frac{\partial_a (\partial_b w_c - \Gamma_{bc}^d w_d)}{\partial_a \partial_b = \partial_b \partial_a} - \Gamma_{ab}^d \nabla_d w_c - \Gamma_{ac}^d \nabla_b w_d = 0$$

3) $\nabla_{[a} R_{bc]}{}^d = 0$

比安基恒等式
(第二比安基)

Bianchi equation

$$\text{pf: } w_e \nabla_a R_{bcd}{}^e = \nabla_a (R_{bcd}{}^e w_e) - R_{bcd}{}^e \nabla_a w_e = \nabla_a ([\nabla_b \nabla_c - \nabla_c \nabla_b] w_d) - R_{bcd}{}^e \nabla_a w_e$$

$$w_e \nabla_{[a} R_{bc]}{}^d = (\nabla_a \nabla_b \nabla_c - \nabla_a \nabla_c \nabla_b) w_d - R_{[bc]d}{}^e \nabla_a w_e = (\nabla_a \nabla_b \nabla_c - \nabla_a \nabla_c \nabla_b) w_d = R_{[abd]}{}^e \nabla_c w_e + R_{[abc]}{}^e \nabla_d w_e = 0$$

有度规 (with metric) $\rightarrow R_{abcd} = g_{de} R_{abc}^e$

4) $R_{abcd} = -R_{abdc}$ (前: 1,2反对称; 3,4反对称)

$\nabla_a g_{bc} = 0 \rightarrow [\nabla_a, \nabla_b] g_{cd} = 0 = R_{abc}^e g_{ed} + R_{abd}^e g_{ce} = R_{abcd} + R_{abdc}$

5) $R_{abcd} = R_{cdab}$ (1,2与3,4对称)

黎曲张量

里奇张量

Ricci tensor

$R_{ac} \equiv R_{abc}^b = R_{abc}^d \cdot g_{d}^b = R_{abcd} g^{bd}$ ($R_{ac} = R_{ca}$)

(6个迹, $g^{ab} R_{abcd} = g^{cd} R_{abcd} = 0$, $g^{ac} R_{abcd} = g^{bd} R_{abcd} = -g^{ad} R_{abcd} = -g^{bc} R_{abcd} = R_{abcd}$ independent only)

里奇标量曲率

Ricci scalar curvature

$R = g^{ac} R_{ac}$

外尔张量

Weyl tensor

(无迹)

$C_{abcd} := R_{abcd} - \frac{2}{n-2} (g_{a[c} R_{d]b} - g_{b[c} R_{d]a}) + \frac{2}{(n-1)(n-2)} R \cdot g_{a[c} g_{d]b}$

1) $C_{abcd} = -C_{bacd} = -C_{abdc} = C_{cdab}$

2) 各种迹为0: $C_{abcd} g^{ab} = 0$, $C_{abcd} g^{ac} = 0$ ($C_{ab}^a = 6$ 个迹)

爱因斯坦张量

Einstein tensor

$\star \nabla^a G_{ab} = 0$

$G_{ab} := R_{ab} - \frac{1}{2} R \cdot g_{ab}$

Proof: $\nabla^a G_{ab} = g^{ac} \nabla_c G_{ab} = \nabla_a R_b^a - \frac{1}{2} \nabla_b R$
 $g^{bd} \times (g^{ae} \times (\nabla_a R_{bcd}^e + \nabla_c R_{abd}^e + \nabla_b R_{cad}^e = 0))$
 $g^{bd} (\nabla_a R_{bcd}^a + (-\nabla_c R_{bd}) + \nabla_b R_{cd} = 0)$
 $\nabla_a R_c^a - \nabla_c R + \nabla_b R_c^b = 0 = 2 \nabla_a R_c^a - \nabla_c R$

由度规到黎曼曲率张量

$g_{ab} \xrightarrow{\nabla_c g_{ab} = 0} \nabla_a \xrightarrow{[\nabla_a, \nabla_b] w_c = R_{abc}^d w_d} R_{abc}^d \xrightarrow{g^{xy} C_{abcd} = 0} C_{abcd}$

$\Gamma_{\mu\nu}^\sigma = \frac{1}{2} g^{\sigma\rho} (g_{\rho\mu,\nu} + g_{\rho\nu,\mu} - g_{\mu\nu,\rho})$

$\nabla_b w_d = \partial_b w_d - \Gamma_{bd}^e w_e$

1) $\nabla_b w_c = \partial_b w_c - \Gamma_{bc}^e w_e$

2) $R_{abc}^d w_d = 2 \nabla_{[a} \nabla_{b]} w_c = 2 (\partial_a - \Gamma_a^e) (\partial_b - \Gamma_b^f) w_c$

$(\partial_a - \Gamma_a^e) (\partial_b - \Gamma_b^f) w_c$

1) $\Gamma_{[ab]}^c = 0$ (对称) 2) $\partial_{[a} \partial_{b]} w_c = 0$ (换序)

反对称, 消

$= -2 w_d \partial_{[a} \Gamma_{b]}^d w_c + 2 w_d \Gamma_{c[a}^e \Gamma_{b]}^d w_e$ (d与e适当交换)

$R_{abc}^d = -2 \partial_{[a} \Gamma_{b]}^d w_c + 2 \Gamma_{c[a}^e \Gamma_{b]}^d w_e$

$$\text{分量: } R_{\mu\nu\sigma}{}^\rho = \Gamma^\rho_{\mu\sigma,\nu} - \Gamma^\rho_{\nu\sigma,\mu} + \Gamma^\lambda_{\sigma\mu}\Gamma^\rho_{\nu\lambda} - \Gamma^\lambda_{\sigma\nu}\Gamma^\rho_{\mu\lambda}$$

$$\text{里奇: } R_{\mu\sigma} = \Gamma^\nu_{\mu\sigma,\nu} - \Gamma^\nu_{\nu\sigma,\mu} + \Gamma^\lambda_{\sigma\mu}\Gamma^\nu_{\nu\lambda} - \Gamma^\lambda_{\sigma\nu}\Gamma^\nu_{\mu\lambda}$$

若 g_{ab} 的 $R_{abc}{}^d = 0$, 则 $\exists g_{\mu\nu}$ 全为常数 (平直时空) 证明略? □

$$\text{选读: } 1) \nabla_a v^a = \partial_a v^a + \Gamma^a_{ab} v^b$$

$$1) \Gamma^\mu_{\mu\sigma} = \frac{1}{2} g^{\mu\lambda} g_{\mu\lambda,\sigma}$$

$$2) g \rightarrow g_{\mu\nu} \text{ 行列式} \rightarrow \frac{\partial g}{\partial g_{\mu\nu}} = g \cdot g^{\mu\nu} \text{ (用余子式)}, g^{\mu\nu} = [g_{\mu\nu}]^{-1} = \frac{A^{\mu\nu}}{g} \text{ 逆矩阵}$$

$$\hookrightarrow \frac{\partial g}{\partial x^\sigma} = \frac{\partial g}{\partial g^{\mu\nu}} \frac{\partial g^{\mu\nu}}{\partial x^\sigma} = g \cdot g^{\mu\nu} \frac{\partial g^{\mu\nu}}{\partial x^\sigma}$$

$$1)+2): \Gamma^\mu_{\mu\nu} = \frac{1}{2g} \frac{\partial g}{\partial x^\nu} = \frac{1}{\sqrt{|g|}} \frac{\partial \sqrt{|g|}}{\partial x^\nu}$$

$$0)+1)+2): \vec{\nabla} \cdot \vec{v} = \nabla_a v^a = v^{\sigma,\sigma} + \frac{v^\sigma}{\sqrt{|g|}} \frac{\partial \sqrt{|g|}}{\partial x^\sigma} = \frac{1}{\sqrt{|g|}} \frac{\partial}{\partial x^\sigma} (\sqrt{|g|} v^\sigma)$$

$$\hookrightarrow \text{笛卡尔: } |g|=1, \vec{\nabla} \cdot \vec{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

$$\text{球: } \sqrt{|g|} = r^2 \sin\theta, \vec{\nabla} \cdot \vec{v} = \frac{1}{r^2 \sin\theta} \left(\frac{\partial (v_r r^2 \sin\theta)}{\partial r} + \frac{\partial (v_\theta r \sin\theta)}{\partial \theta} + \frac{\partial (v_\varphi r \sin\theta)}{\partial \varphi} \right)$$

$$= \frac{1}{r^2} \frac{\partial (r^2 v_r)}{\partial r} + \frac{1}{r \sin\theta} \frac{\partial (v_\theta \sin\theta)}{\partial \theta} + \frac{1}{r \sin\theta} \frac{\partial v_\varphi}{\partial \varphi}$$

$$e_r = (\partial/\partial r), e_\theta = \frac{1}{r} (\partial/\partial \theta), e_\varphi = \frac{1}{r \sin\theta} (\partial/\partial \varphi)$$

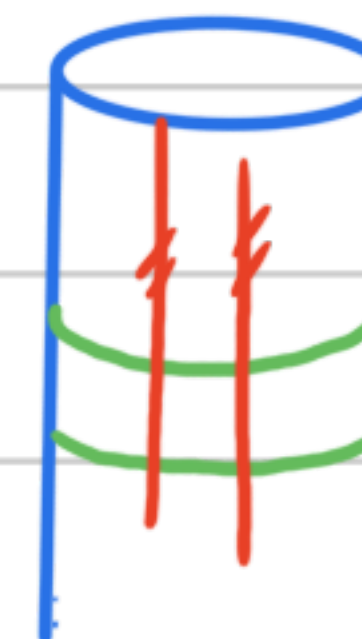
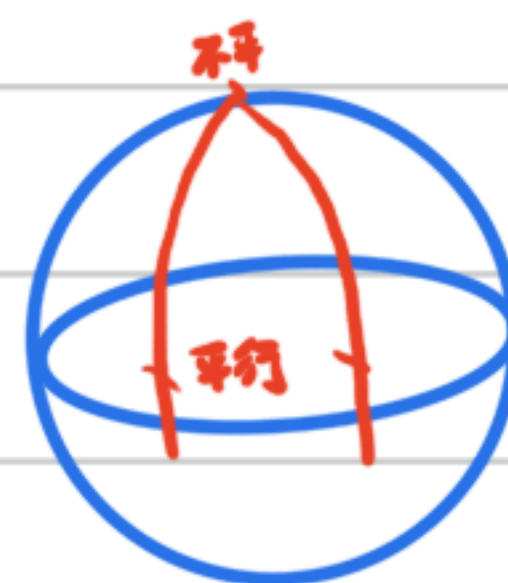
内禀曲率, 外曲率 (把 n 维流形嵌入到 $n+1$ 维定义的曲率)

intrinsic

黎曼曲率为内禀曲率 1) 导数算符非对易: $\nabla_{[a} \nabla_{b]} w_c = R_{abc}{}^d w_d$

2) 矢量平移曲线依赖性:

3) 存在初始平行后来不平行的测地线



内禀曲率 = 0 (可用平面围)
外曲率 ≠ 0

一图流

